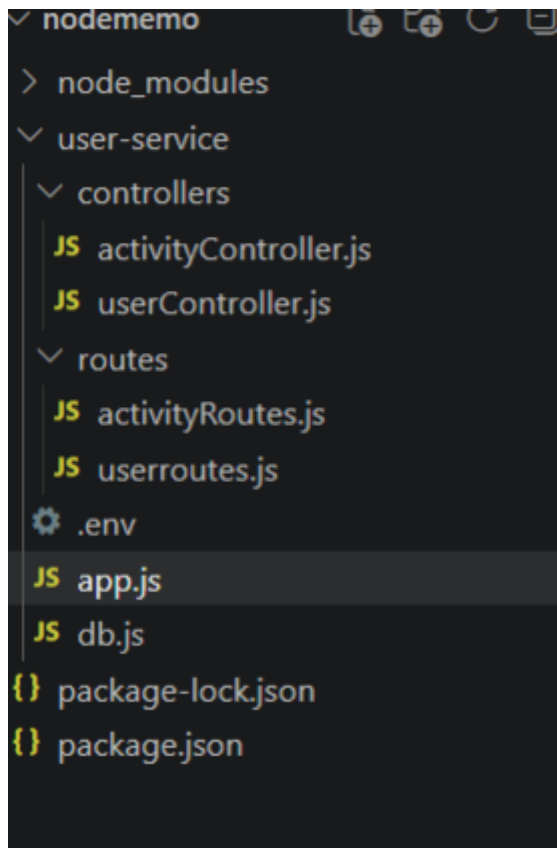


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Practical 7



App.js:

```
1  const dns = require("dns");
2  dns.setServers(["8.8.8.8", "8.8.4.4"]);
3
4  require("dotenv").config();
5
6  const express = require("express");
7  const app = express();
8
9  require("./db");
10
11 app.use(express.json());
12
13 const activityRoutes = require("./routes/activityRoutes");
14 const userRoutes = require("./routes/userRoutes");
```

```
app.use("/", activityRoutes);
app.use("/", userRoutes);

app.listen(5001, () => {
  console.log("User service running on port 5001");
});
```

DB.js

```
const mongoose = require('mongoose');

const MONGO_URI = process.env.MONGO_URI;

mongoose.connect(MONGO_URI, {
  useNewUrlParser: true,
  useUnifiedTopology: true
}).then(() => console.log("MongoDB Atlas connected"))
.catch((err) => console.error("MongoDB connection error"));

const userSchema = new mongoose.Schema({
  username: {
    type: String,
    required: true
  },
  email: {
    type: String,
```

```
14    type: String,
15    required: true
16  },
17  profile: {
18    type: mongoose.Schema.Types.Mixed,
19    default: {}
20  }
21 });
22
23
24 const activitySchema = new mongoose.Schema({
25   user_id: {
26     type: mongoose.Schema.Types.ObjectId,
27     required: true
```

```
27     required: true,  
28     ref: "User"  
29   },  
30   action: {  
31     type: String,  
32     required: true  
33   },  
34   timestamp: {  
35     type: Date,  
36     default: Date.now  
37   }  
38 });
```

```
1  const express = require("express");  
2  const router = express.Router();  
3  
4  const {  
5    createUser,  
6    getUser  
7  } = require("../controllers/userController");  
8  
9  router.post("/users", createUser);  
10 router.get("/users/:id", getUser);  
11  
12 module.exports = router;
```

```
38 });  
39  
40 const User = mongoose.model("User", userSchema);  
41 const Activity = mongoose.model("Activity", activitySchema);  
42  
43 module.exports = { User, Activity };
```

```
38     res.json(user);  
39  
40   } catch (error) {  
41     res.status(500).json({  
42       message: "Error getting user",  
43       error: error.message  
44     });  
45   }  
46 };  
47  
48 module.exports = {  
49   createUser,  
50   getUser  
51 };
```

```

1  const mongoose = require( mongoose );
2  const { Activity } = require("../db");
3
4  // POST - Create Activity
5  const createActivity = async (req, res) => {
6    try {
7      const { user_id, action } = req.body;
8
9      if (!mongoose.Types.ObjectId.isValid(user_id)) {
10         return res.status(400).json({
11           message: "Invalid user_id. It must be a valid Mongol
12         });
13       }
14
15       const activity = await Activity.create({

```

```

17         action
18       });
19
20       res.status(201).json({
21         message: "Activity created successfully",
22         activity
23       });
24
25     } catch (error) {
26       res.status(500).json({
27         message: "Error creating activity",
28         error: error.message

```

```

38     if (!mongoose.Types.ObjectId.isValid(user_id)) {
39         return res.status(400).json({
40             message: "Invalid user_id. It must be a valid Mongo
41         });
42     }
43
44     const activities = await Activity.find({
45         user_id
46     }).sort({ timestamp: -1 });
47
48     res.json(activities);
49
50 } catch (error) {

```

Outputs:

```

PS C:\Users\Geetha\OneDrive\Documents\nodememo> Invoke-WebRequest -Uri "http://localhost:5001/users"
>> -Method POST
>> -ContentType "application/json"
>> -Body '{"username":"Divya","email":"divya@gmail.com","profile":{}}'

message          user
-----
User created successfully @{username=Divya; email=divya@gmail.com; _id=6ab4ef3ddb100d25078d783b; __v=0}

```

```

message          user
-----
User created successfully @{username=Divya; email=divya@gmail.com; _id=6ab4ef3ddb100d25078d783b; __v=0}

PS C:\Users\Geetha\OneDrive\Documents\nodememo>

```

Pretty-print 

```

{
  "_id": "6ab4ef3ddb100d25078d783b",
  "username": "Divya",
  "email": "divya@gmail.com",
  "__v": 0
}

```

```
{
  "_id": "6ab4f17adb100d25078d783c",
  "user_id": "6ab4ef3ddb100d25078d783b",
  "action": "SEARCH_PYTHON",
  "timestamp": "2026-09-24T09:46:34.473Z",
  "__v": 0
}
```

```
PS C:\Users\Geetha\OneDrive\Documents\nodememo> Invoke-RestMethod -Uri "http://localhost:5001/activities" `
>> -Method POST `
>> -ContentType "application/json" `
>> -Body '{"user_id":"6ab4ef3ddb100d25078d783b","action":"SEARCH_PYTHON"}'

message          activity
-----
Activity created successfully @{user_id=6ab4ef3ddb100d25078d783b; action=SEARCH_PYTHON; _id=6ab4f17adb100d25078...
```